

Ajay Rajendra Kumar

RESEARCH ASSISTANT · VISION-LANGUAGE MODELS

Boston, MA, USA

✉ rajendrakumar [dot] a [at] northeastern [dot] edu | 🏠 ajaystar8.github.io | 📧 ajaystar8 | 🌐 ajayr-kumar | 📄 Google Scholar

“Towards Generalizable and Reliable Artificial Intelligence”

Summary

Computer science graduate conducting research in computer vision and multimodal learning, with interests in developing models capable of reliable perception and reasoning over multimodal inputs in long-horizon settings. My research focuses on agentic architectures for active perception, evidence retrieval, and improving spatial and causal reasoning capabilities. Currently a research assistant in Prof. Lorenzo Torresani's lab at Northeastern University; previously worked with Prof. Huaizu Jiang's Visual Intelligence Lab.

Field of Interest

MY RESEARCH INTERESTS INCLUDE, BUT ARE NOT CONFINED TO:

- Long-horizon multimodal reasoning.
- Grounded multimodal learning and evidence retrieval.
- Agentic architectures for active perception.
- Continual learning and memory for MLLMs.
- 3D scene understanding and dense visual SLAM.
- Efficient deep learning for real-time perception.

Education

Northeastern University

Boston, USA

M.S. IN COMPUTER SCIENCE

Sep. 2024 - May 2026

- GPA: 4.0/4.0
- Capstone: “Dense Visual SLAM on Edge Devices”
- Advised by Prof. Huaizu Jiang

Manipal Institute of Technology

Manipal, India

B.TECH. IN COMPUTER SCIENCE

Aug. 2020 - Jul. 2024

- GPA: 9.3/10.0
- Minor: Computational Intelligence
- Dissertation: “Evaluating Deep Learning Approaches for Fracture Segmentation in X-ray Images”
- Advised by Prof. Subhamoy Mandal

Work Experience

Research Assistant (Advisor: Prof. Lorenzo Torresani)

Boston, USA

NORTHEASTERN UNIVERSITY, KHOURY COLLEGE OF COMPUTER SCIENCES

Nov. 2025 - Present

- Improved video question answering performance of the PerceptionLM-8B by 12% on the EgoBlind benchmark through prompt optimization and supervised fine-tuning.
- Engineered an agentic pseudo chain-of-thought generation pipeline integrating SAM3, Qwen-2.5-VL-7B, and PerceptionLM-8B to automatically generate grounded visual concepts and textual evidence, augmenting the Open-o3-Video dataset with 10K+ high-quality training samples.
- Trained Qwen-2.5-VL-7B using a two-stage curriculum comprising LoRA-based supervised fine-tuning (SFT) followed by GRPO reinforcement learning (RL) on the synthesized grounded supervision dataset.
- Achieved a 5% improvement on the V-STaR video benchmark, with consistent gains across other benchmarks; currently extending the approach to long-horizon multimodal reasoning via audio-video evidence retrieval and memory-augmented reasoning.

Research Assistant (Advisor: Prof. Huaizu Jiang)

Boston, USA

NORTHEASTERN UNIVERSITY, KHOURY COLLEGE OF COMPUTER SCIENCES

Apr. 2025 - Mar. 2026

- Designed and trained a lightweight dense feature matching network to estimate pixel-wise correspondence confidence, filtering unreliable matches to improve the robustness and accuracy of downstream camera pose estimation.
- Architected a scalable Kubric-based synthetic data generation pipeline to render 2K+ stereo video sequences with diverse camera trajectories, object configurations, and HDRI environments for training the confidence estimation network.
- Co-developed NeuSLAM, a dense visual SLAM system integrating the confidence-guided front-end with an efficient classical pose-tracking back-end, achieving state-of-the-art F1-all of 9.9% on KITTI 2015 optical flow and competitive stereo disparity on KITTI 2015 and ETH3D in zero-shot synthetic-to-real evaluation.
- Contributed to the deployment, optimization, and experimental validation of NeuSLAM on Jetson Orin Nano-based robotic platforms, achieving trajectory accuracy competitive with DROID-SLAM at 6.5× faster speed and 87% lower GPU memory on TUM RGB-D and FLSea-VI datasets.

Research Assistant (Advisors: Prof. Anna Lamin & Prof. Valentina Marano)

NORTHEASTERN UNIVERSITY, D'AMORE-MCKIM SCHOOL OF BUSINESS

- Engineered an end-to-end data acquisition pipeline to scrape and preprocess CSR disclosures from Indian companies, curating a dataset of 100K+ records spanning five years for downstream empirical analysis.
- Built scalable Python-based ETL pipelines for JSON and PDF ingestion, incorporating structure-aware document chunking, metadata extraction, and automated preprocessing to support retrieval and information extraction workflows.
- Architected an LLM-powered document intelligence pipeline combining semantic retrieval, targeted OCR, and OpenAI models to extract, clean, normalize, and transform unstructured reports into structured tabular datasets for downstream analytics.
- Scored 10K+ Crunchbase company profiles using a two-stage LLM pipeline (embedding retrieval + GPT-4.1 Likert scoring) against ten research criteria, enabling empirical analysis of domestic market orientation of Indian firms.

Boston, USA

Feb. 2025 - May 2026

Software Development Intern

FIDELITY INVESTMENTS

- Reduced dashboard query latency by 90% by migrating the underlying data warehouse from Oracle to Snowflake and re-architecting query execution for large-scale change-ticket analytics.
- Delivered full-stack features for an internal change-ticket management platform using Angular and Spring Boot, redesigning core workflows for creating, tracking, and managing engineering change requests.
- Designed and integrated RESTful APIs using Spring Boot to connect the Angular frontend with Snowflake-backed backend services, enabling efficient ticket creation, tracking, and reporting workflows.

Bengaluru, India

Jun. 2023 - Aug. 2023

Selected Projects

Pose Graph Sparsification by Maximizing Algebraic Connectivity

MOBILE ROBOTICS

- Formulated and implemented an iterative variant of the algebraic connectivity maximization algorithm for pose graph sparsification.
- Evaluated on benchmark pose graphs using GTSAM and SE-Sync, achieving sparser graphs with minimal performance loss.
- Keywords: Pose Graph Optimization, Graph Sparsification, SLAM, GTSAM, SE-Sync

Northeastern University, Boston, USA

Mar. 2025

Hallucination Detection in Large Language Models

NATURAL LANGUAGE PROCESSING

- Evaluated hallucination detection methods from “Joint Evaluation of Answer and Reasoning Consistency for Hallucination Detection in Large Reasoning Models” (AAAI 2026) across multiple open-source LLMs.
- Identified limited generalization of the proposed methods beyond the original evaluated models through thorough experiments.
- Keywords: Hallucination Detection, Large Language Models, PyTorch, Python

Northeastern University, Boston, USA

Mar. 2025

Improving Text-Driven Human Motion Generation through Equivariance

DEEP LEARNING

- Integrated equivariant neural networks into motion diffusion models for human motion generation.
- Improved matching accuracy and achieved competitive FID scores against existing baselines.
- Keywords: Human Motion Generation, Diffusion Models, Equivariant Neural Networks, Motion Diffusion, PyTorch

Northeastern University, Boston, USA

Mar. 2025

Rethinking Depth for Object Detection with YOLOv3

COMPUTER VISION

- Implemented YOLOv3 and RGB-D variants to study the impact of depth cues on object detection.
- Observed limited performance gains, indicating need for improved depth fusion architectures.
- Keywords: Object Detection, RGB-D Vision, Depth Fusion, YOLOv3, Multimodal Perception, OpenCV, Python

Northeastern University, Boston, USA

Dec. 2024

Addressing Vaccine Misinformation on Social Media

RESEARCH ASSISTANT (BACHELOR’S)

- Fine-tuned transformer models for misinformation classification on social media data.
- Developed similarity-based method for anomalous user detection, achieving strong classification performance.
- Keywords: Misinformation Detection, Social Media Analysis, Deep Learning, PyTorch

Manipal Institute of Technology, Manipal, India

Mar. 2024

Publications

Underline denotes co-first authors or co-corresponding authors

INTERNATIONAL CONFERENCE

1. Aniket Gupta, Tianye Ding, **Ajay Rajendra Kumar**, Dennis Giaya, Pia Bideau, Charles Saunders, Qu Cao, Aruni

RoyChowdhury, Hanumant Singh, and Huaizu Jiang, “NeuSLAM: Dense Visual SLAM on Edge Devices,” accepted to *ICRA 2026 Workshop on From Sea to Space: Advancing Perception in Harsh Domains*; accepted to *CVPR 2026 Demo Track*.

2. C. Rao*, G. M. Prabhu*, **Ajay Rajendra Kumar***, S. Gupta, and N. P. Shetty, “Addressing Vaccine Misinformation on Social Media by Leveraging Transformers and User Association Dynamics,” in *Int. Conf. Mach. Learn. Data Eng. (ICMLDE)*, 2023.

Teaching Experience

Object-Oriented Design (CS2163)

*Manipal Institute of Technology,
Manipal, India*

UNDERGRADUATE TEACHING ASSISTANT (INSTRUCTOR: PROF. ANUP BHAT)

Aug. 2023 - Dec. 2023

- Assisted students with object-oriented programming concepts, software design principles, and Java-based programming assignments.
- Conducted office hours and supported evaluation of programming assignments.

Problem Solving Using Computers (CS1081)

*Manipal Institute of Technology,
Manipal, India*

UNDERGRADUATE TEACHING ASSISTANT (INSTRUCTOR: PROF. T. SUJITHRA)

Aug. 2023 - Dec. 2023

- Guided students in programming fundamentals, algorithmic problem solving, and debugging techniques.
- Assisted with lab sessions and assignment evaluations for undergraduate students.

Indian Society for Technical Education (ISTE), Manipal Chapter

*Manipal Institute of Technology,
Manipal, India*

MACHINE LEARNING HEAD

Jun. 2022 - Dec. 2023

- Led machine learning initiatives, organizing technical workshops and mentoring students on ML projects.
- Coordinated hands-on sessions covering machine learning concepts and practical implementations.

Honors & Awards

Mar. 2025 **Most Innovative Project (among 40+ submissions)**, PhD-level Deep Learning Course (CS7150), Northeastern University

Boston, USA

Oct. 2023 **Undergraduate Sponsorship Grant**, Manipal Academy of Higher Education (MAHE)

Dehradun, India

Aug. 2022 **National 2nd Runner-Up**, Daimler India Commercial Vehicles Hackathon

India

Apr. 2022 **1st Runner-Up**, Transfusion Medicine Hackathon

Manipal, India

Jun. 2021 **Winner**, Space Odyssey Web Development Hackathon

Manipal, India

Jan. 2021 **Scholar's Scholarship Award**, Manipal Academy of Higher Education (MAHE)

Manipal, India